**ESE381 Embedded Microprocessor Systems Design II**

Spring 2026, K. Short
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## Laboratory 06: AVR128DB48 I2C Module with Sparkfun 20 x 4 SerLCD and LM75 Temperature Sensor

To be performed the week starting March 8th.

### Prerequisite Reading

1 AVR128DB48 Data Sheet Section 29, TWI - Two-Wire Interface
2. UM10204 I2C-Bus Specification and User Manual
3. Sparkfun 20 x 4 SerLCD https://learn.sparkfun.com/tutorials/avr-based-serial-enabled-lcds-hookup-guide
4. LM75 Digital Temperature Sensor and Thermal Watchdog with 2-Wire Interface Data Sheet.
5. LabLec 06: LM75 I2C Interface and Basic I2C Transactions
6. Print Floating Point Numbers in AVR C with Atmel Studio
https://startingelectronics.org/articles/atmel-AVR-8-bit/print-float-atmel-studio-7/

### Task List

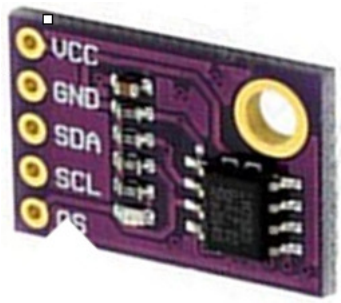
***Task 1: Drivers to Write SparkFun 20 x 4 SerLCD via I2C (TWI)***
***Task 2: Using TWI0 Module to Provide an I2C Interface to the LM75***
***Task 3: Program to Read Temperature from an LM75***
***Task 4: LM75 Temperature Measuring System***

### Overview

This laboratory focuses on the I2C serial protocol. The modules in the AVR128DB48 that provide the hardware to implement this protocol are referred to in the AVR128DB48 data sheet as **two-wire interface (TWI) modules**. In the previous two laboratories you have interfaced the SerLCD to the AVR128DB48 using the asynchronous serial protocol and the serial peripheral interface (SPI) protocol. In this laboratory you will interface the SerLCD to the AVR128DB48 using the I2C protocol. The display functionality that you will implement is essentially the same as in the previous laboratory.

You will also interface to an LM75 temperature sensor that also uses the I2C protocol. The LM75 includes a silicon band gap temperature sensor, a delta-sigma analog-to-digital converter, and a digital overtemperature detector. A microcontroller can query the LM75 through its I2C interface to read the sensor's temperature at any time. You will use an LM75 mounted on a breakout board,

as shown below.




The breakout board has been hardwired so that the least significant three bits of the LM75's slave address are 000. The slave address control byte in the LM75 is shown below.

| BIT 7 | BIT 6 | BIT 5 | BIT 4 | BIT 3 | BIT 2 | BIT 1 | BIT 0 |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | A2 | A1 | A0 | R/$\overline{W}$ |

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The LM75 has four registers.

| REGISTER NAME | ADDRESS (hex) | POR STATE (hex) | POR STATE (binary) | POR STATE (°C) | READ/ WRITE |
|---|---|---|---|---|---|
| Temperature | 00 | 000X | 0000 0000 0XXX XXXX | — | Read only |
| Configuration | 01 | 00 | 0000 0000 | — | R/W |
| $T_{HYST}$ | 02 | 4B0X | 0100 1011 0XXX XXXX | 75 | R/W |
| $T_{OS}$ | 03 | 500X | 0101 0000 0XXX XXXX | 80 | R/W |

*X = Don't care.*

To, keep this laboratory simple, we will only be using the Temperature register. Registers are accessed using a pointer to the register. This pointer value defaults to point to the Temperature register when the LM75 is reset. The most significant nine bits of the 16-bit Temperature register hold the measured temperature as a 2's complement number.

| UPPER BYTE | | | | | | | | LOWER BYTE | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| Sign bit<br>1= Negative<br>0 = Positive | MSB<br>64˚C | 32°C | 16°C | 8°C | 4°C | 2°C | 1°C | LSB<br>0.5°C | X | X | X | X | X | X | X |

*X = Don't care.*

One of the tasks of this laboratory is to design and debug an I2C interface to the LM75 and write simple drivers to configure the AVR128DB48s TWI0 module to read the LM75's temperature. These tasks will help you become familiar with using the AVR128DB48's TWI module to implement a simple I2C protocol interface.

**Design Tasks**

***Design Task 1: Drivers to Write SparkFun 20 x 4 SerLCD via TWI (I2C)***
Draw a schematic of the interconnection of the SerLCD to the AVR128DB48 using TWI and the SerLCD's I2C pins. On the SerLCD the connections are SDA and SCL. On the AVR128DB48 use TWI module 0 (TWI0). On the AVR128DB48 Curiosity Nano board, SDA must be driven by PA2 and SCL must be driven by PA3. PA2 and PA3 are the default SDA and SCL signals, respectively, for TWI0.

In the "datasheet" for the SerLCD, under Commands, is the following:

Ctrl+y / 25 / 0x19 - Change the TWI address. Follow Ctrl+x with number 0 to 255. 114 (0x72) is default. We can use the default address of 0x72, so you will not need to write this command to the SerLCD.

For TWI serial clock speed use standard mode (100kHz). Once you have this working you can increase the speed to the High Speed Mode of 400kHz, or as close as you can get to that speed, if you want.

You must write a function to properly configure TWI0 to communicate with the SerLCD. You must write a second function that sends a character to the SerLCD.

The required function templates are:
To configure TWI0:

```
void init_twi0_SerLCD (void);
```

To write a character using TWI0:

```
int write_twi0_SerLCD(uint8_t saddr, uint8_t data);
```

You must also create an update SerLCD function.

```
void update_SerLCD(void);
```

Write a program named `write_SerLCD_image_TWI0` that uses the `update_SerLCD` function along with the above functions and the `sprintf` function to copy the contents of the image buffer in the AVR128DB48's SRAM to the SerLCD..

**Submit your schematic and C source file for this program as part of your prelab.**

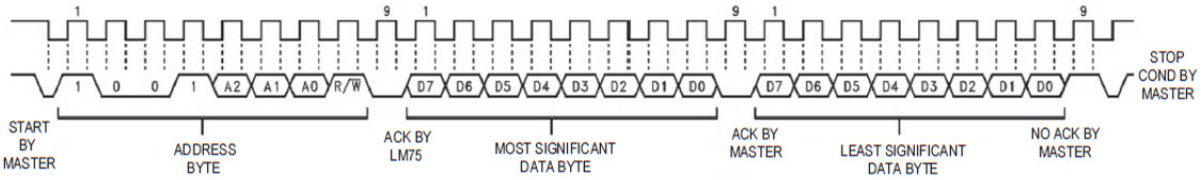
***Design Task 2: Program to Read the Temperature from an LM75***

Study the data sheet for the LM75. This device has only an I2C interface. The AVR128DB48's TWI0 module must be configured to use this interface to communicate with the LM75. SDA must be driven by PA2 and SCL must be driven by PA3. PA2 and PA3 are the default SDA and SCL signals, respectively, for TWI0.

Draw a schematic diagram showing the wiring of the LM75 and its interface to the AVR128DB48. **Submit this item as part of your prelab.**

**Write a function** named `TWI0_LM75_init` that is passed no arguments and returns no value. This function initializes the AVR128DB48's TWI0 module to communicate with the LM75. The bit transfer rate between the AVR128DB48 and the LM75 must be as fast as possible, but less than or equal to 400 kb/s.

The overall objective of this task is just to read the temperature measured by the LM75. The I2C waveform to perform a two Data byte read of the LM75 follows.




(a) TYPICAL 2-BYTE READ FROM PRESET POINTER LOCATION SUCH AS TEMP, $T_{HIGH}$, $T_{LOW}$.

**Write a function** named `TWIO_LM75_read` that is passed one `uint8_t` argument. This argument's value is the LM75's 7-bit slave address and the R/$\overline{W}$ bit. This function returns a `uint16_t` that is the contents of the Temperature register with the don't care bits forced to 0s.

**Write a program** named `read_LM75_temp_test`. Your program must use the previously defined functions to read the Temperature register and compute the temperature. Use the following global variables to store the values read from the LM75's Temperature register and computed temperature.

```
uint8_t temp_reg_high;  //high byte of LM75 Temperature register
uint8_t temp_reg_low;   //low byte of LM75 Temperature register
uint16_t LM75_temp_reg = 0; //LM75 Temperature register contents
int16_t LM75_temp = 0;  //right adjusted 2's complement LM75 temp
```

Your program does not need to display the above information on the SerLCD. It is sufficient for verification purposes to display the data in a watch window.

**Submit your LM75 schematic and program listing as part of your prelab. Make sure that your program includes program headers, function headers, and clear comments throughout.**

***Design Task 3: Program to Read and Display the Temperature from an LM75***

**Write a program** named `display_LM75_temp`. This program is an extension of the program of Task 2. It continually displays the LM75's temperature as a signed decimal value on line 1 of the LCD. Provide an appropriate prefix for the value displayed on the LCD. The program should display the temperature with a resolution of +/- 0.5 °C.

The temperature range of the LM75 is +125 to -55 °C. You can use the `sprintf` function to put a formatted string that displays the temperature into the SerLCD SRAM buffer for line 1 of the display. However, if you want to use floating point specifiers in the format string of your `sprintf`, you must change the Studio 7 linker settings in your project. The default settings in Atmel Studio 7 disable floating point for `sprintf` / `printf` type functions to save microcontroller flash memory. The prerequisite reading document *Print Floating Point Numbers in AVR C with Atmel Studio 7* tells you how to properly change Atmel Studio 7's linker settings.

Use the watch window to verify that your readings from the LM75 and computed temperature are correct.

***Design Task 4: Temperature Measuring System***

Use your work from Tasks 1, 2, and 3, along with any other necessary functions to create a systems that measures and displays ambient temperature. Write a program named `temp_meas_LM75` that continuously measures and displays temperature in °C and °F.

***Submit your source code as part of your prelab.***

Watch 1

| Name | Value | Type |
|---|---|---|
| temp_reg_high | 0x17 | uint8_t{data}@0x4026 |
| temp_reg_low | 0x40 | uint8_t{data}@0x4025 |
| LM75_temp_reg | 0x1700 | uint16_t{data}@0x400e |
| LM75_temp | 0x002e | int16_t{data}@0x400c |
| dsp_buff1,ascii | 0x4010 | char[21]{data}@0x4010 |
| [0],ascii | 0x54 'T' | char{data}@0x4010 |
| [1],ascii | 0x65 'e' | char{data}@0x4011 |
| [2],ascii | 0x6d 'm' | char{data}@0x4012 |
| [3],ascii | 0x70 'p' | char{data}@0x4013 |
| [4],ascii | 0x20 ' ' | char{data}@0x4014 |
| [5],ascii | 0x3d '=' | char{data}@0x4015 |
| [6],ascii | 0x20 ' ' | char{data}@0x4016 |
| [7],ascii | 0x32 '2' | char{data}@0x4017 |
| [8],ascii | 0x33 '3' | char{data}@0x4018 |
| [9],ascii | 0x2e '.' | char{data}@0x4019 |
| [10],ascii | 0x30 '0' | char{data}@0x401a |
| [11],ascii | 0x0 ' | char{data}@0x401b |
| [12],ascii | 0x0 ' | char{data}@0x401c |
| [13],ascii | 0x0 ' | char{data}@0x401d |
| [14],ascii | 0x0 ' | char{data}@0x401e |
| [15],ascii | 0x0 ' | char{data}@0x401f |
| [16],ascii | 0x0 ' | char{data}@0x4020 |
| [17],ascii | 0x0 ' | char{data}@0x4021 |
| [18],ascii | 0x0 ' | char{data}@0x4022 |
| [19],ascii | 0x0 ' | char{data}@0x4023 |
| [20],ascii | 0x0 ' | char{data}@0x4024 |

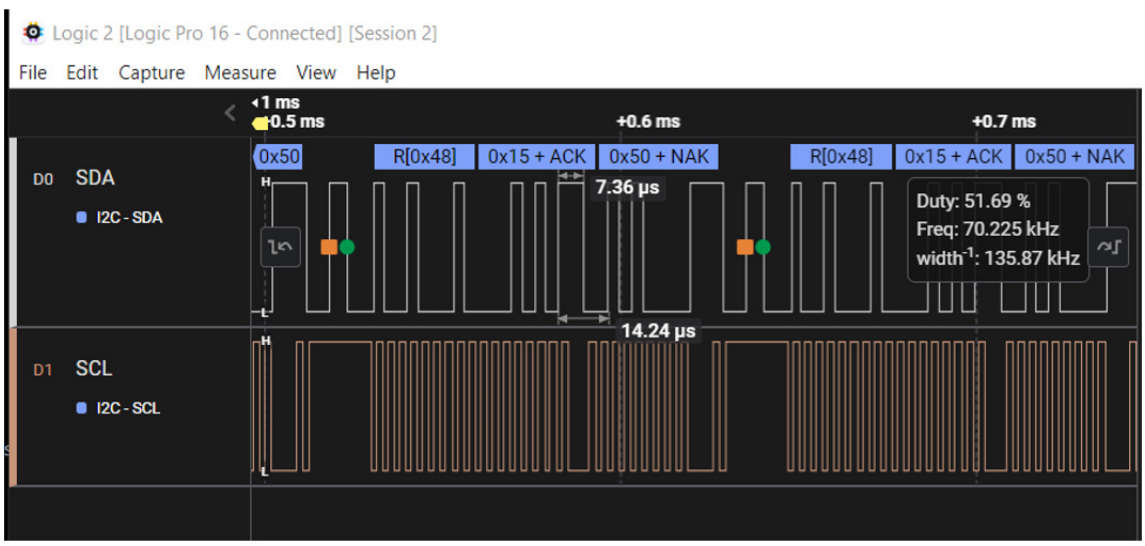
## Laboratory Activities

***Laboratory Task 1: Drivers to Write SparkFun 20 x 4 SerLCD via TWI (I2C)***
Create a project named `SerLCD_write_TWI0_test`. Add your program `SerLCD_write_TWI0_test` and compile it. Connect and configure the Saleae Logic Analyzer so that you can view the transactions with the SerLCD on the I2C bus.

In this and future laboratories you must have the Saleae logic analyzer connected to the AVR128DB48 and showing the appropriate waveforms for any task involving a serial protocol to get the signoff for the task.

**Have a TA verify the correct operation of your system and the correct waveforms on the Saleae Logic Analyzer and get the TA's signature.**

***Laboratory Task 2: Program to Read the Temperature from an LM75***
Create a project named `read_LM75_temp`. Add your program `read_LM75_temp` and compile it. Connect and configure the Saleae Logic Analyzer so that you can view the transactions on the I2C bus.




Run your program and verify that it operates properly. Verify that your readings of the LM75's Temperature register and your computation of the temperature are correct by displaying the appropriate variables in the Watch window.

Watch 1

| Name | Value | Type |
|---|---|---|
| temp_reg_high | 0x14 | uint8_t{data}@0x4005 |
| temp_reg_low | 0x70 | uint8_t{data}@0x4004 |
| LM75_temp_reg | 0x1400 | uint16_t{data}@0x4002 |
| LM75_temp | 0x0028 | int16_t{data}@0x4000 |

If there are problems, identify their sources and correct them. Carefully use a "cold can" and heat

gun to provide a range of temperatures to your LM75.

**Have a TA verify the correct operation of your system and the correct waveforms on the Saleae Logic Analyzer and get the TA's signature.**

***Laboratory Task 3: Program to Read and Display the Temperature from an LM75***
Create a project named `display_LM75_temp`. Add your program `display_LM75_temp` program and compile it. Run your program and verify that it operates properly. Your program must display both positive and negative temperatures correctly on the SerLCD. Show a TA your Watch window showing the appropriate values for the variables and line 1 LCD buffer. Show that what is displayed on the LCD is consistent with the variable values.

Watch 1

| Name | Value | Type |
|---|---|---|
| temp_reg_high | 0x15 | uint8_t{data}@0x4024 |
| temp_reg_low | 0x90 | uint8_t{data}@0x4023 |
| LM75_temp_reg | 0x1580 | uint16_t{data}@0x4010 |
| LM75_temp | 0x002b | int16_t{data}@0x400e |
| dsp_buff1 | 0x4012 | char[17]{data}@0x4012 |
| [0] | 0x54 | char{data}@0x4012 |
| [1] | 0x65 | char{data}@0x4013 |
| [2] | 0x6d | char{data}@0x4014 |
| [3] | 0x70 | char{data}@0x4015 |
| [4] | 0x20 | char{data}@0x4016 |
| [5] | 0x3d | char{data}@0x4017 |
| [6] | 0x20 | char{data}@0x4018 |
| [7] | 0x20 | char{data}@0x4019 |
| [8] | 0x20 | char{data}@0x401a |
| [9] | 0x32 | char{data}@0x401b |
| [10] | 0x31 | char{data}@0x401c |
| [11] | 0x2e | char{data}@0x401d |
| [12] | 0x35 | char{data}@0x401e |
| [13] | 0x00 | char{data}@0x401f |
| [14] | 0x00 | char{data}@0x4020 |
| [15] | 0x00 | char{data}@0x4021 |
| [16] | 0x00 | char{data}@0x4022 |

**Have a TA verify the correct operation of your system and get the TA's signature.**

***Laboratory Task 4: Using a UART to Display Messages on the SerLCD***
Create a project named `temp_meas_LM75`.

Load, debug, and run your `temp_meas_LM75` program.

**When your program is working correctly, have a TA verify that your program performs as required. Get the TA's signature.**

**Leave the circuit you have constructed on your breadboard, it may be used in later laboratories.**

## Questions

1. What is the maximum `SCL` clock frequency for the LM75? Explain how you determined it. At

what `SCL` frequency are you operating your LM75, explain how you know that.

2. What is the value to be loaded into the MBAUD register to get the desired maximum `SCL` clock frequency. Show your calculations.

3. What value should the DBGRUN bit have in the DBGCTRL register when you are debugging your software and why?

4. When viewing SDA and SCL on the Saleae Logic Analyzer, what do the red square and green circle represent?

5. In one short sentence, explain whether the master or the slave should be providing the acknowledge after a byte transfer.